Standard Costs and variance Analysis

Chapter VI

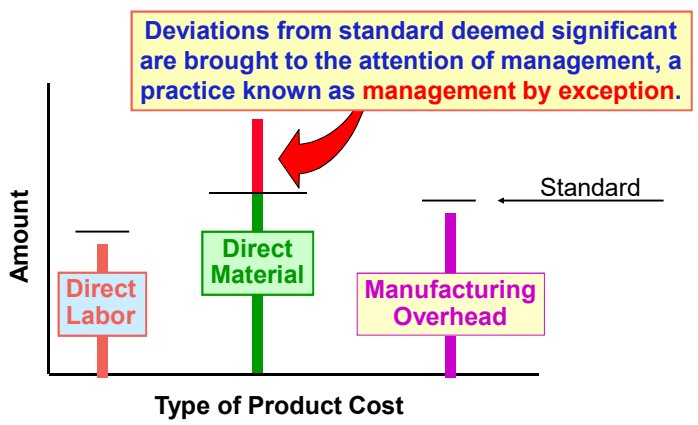
Standard Costs

Standards are benchmarks or "norms" for measuring performance. Two types of standards are commonly used.

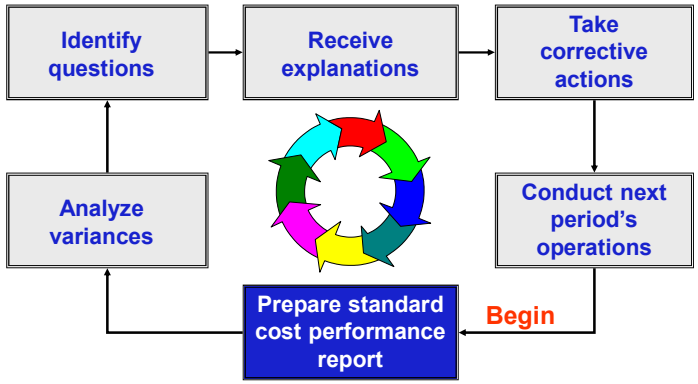
Quantity standards specify how much of an input should be used to make a product or provide a service.

Cost (price) standards specify how much should be paid for each unit of the input.

Standard Costs



Variance Analysis Cycle



Setting Standard Costs

Accountants, engineers, purchasing agents, and production managers combine efforts to set standards that encourage efficient future production.

Setting Standard Costs

Should we use ideal standards that require employees to work at 100 percent peak efficiency?

I recommend using practical standards that are currently attainable with reasonable and efficient effort.

Engineer

Managerial Accountant

Setting Direct Material Standards

Price Standards

Final, delivered cost of materials, net of discounts.

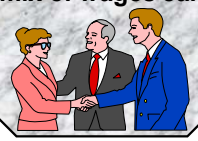
Quantity Standards

Summarized in a Bill of Materials.

Setting Direct Labor Standards

Rate Standards

Often a single rate is used that reflects the mix of wages earned.



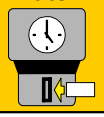
Time Standards

Use time and motion studies for each labor operation.

Setting Variable Overhead Standards

Rate Standards

The rate is the variable portion of the predetermined overhead rate.



Activity Standards

The activity is the base used to calculate the predetermined overhead.

Standard Cost Card – Variable Production Cost

A standard cost card for one unit of product might look like this:

Inputs	A Standard Quantity or Hours	B Standard Price or Rate	A x B Standard Cost per Unit
Direct materials	3.0 lbs.	$ 4.00 per lb.	$ 12.00
Direct labor	2.5 hours	14.00 per hour	35.00
Variable mfg. overhead	2.5 hours	3.00 per hour	7.50
Total standard unit cost			$ 54.50

Standards vs. Budgets

Are standards the same as budgets?

A budget is set for total costs.

A standard is a per unit cost.

Standards are often used when preparing budgets.

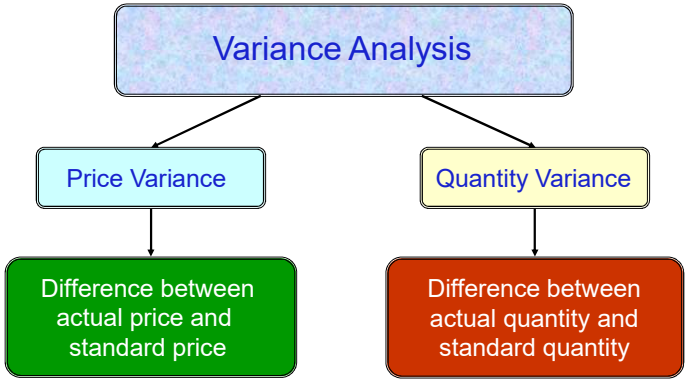
Price and Quantity Standards

Price and and quantity standards are determined separately for two reasons:

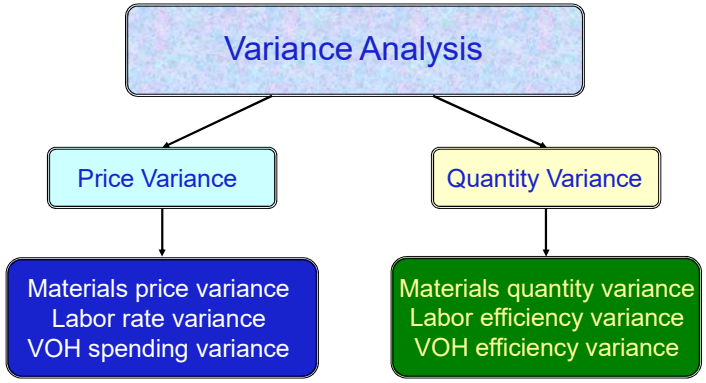
❶ The purchasing manager is responsible for raw material purchase prices and the production manager is responsible for the quantity of raw material used.

❷ The buying and using activities occur at different times. Raw material purchases may be held in inventory for a period of time before being used in production.

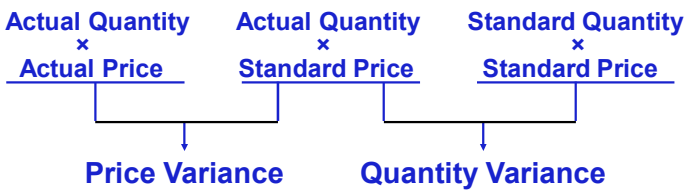
A General Model for Variance Analysis



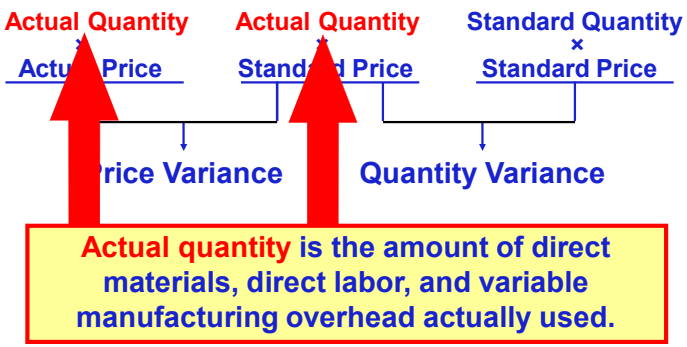
A General Model for Variance Analysis



A General Model for Variance Analysis



A General Model for Variance Analysis



A General Model for Variance Analysis

Actual Quantity × Actual Price | Actual Quantity × Standard Price | Standard Quantity × Standard Price

Price Variance | Quantity Variance

Standard quantity is the standard quantity allowed for the actual output for the period.

A General Model for Variance Analysis

Actual Quantity × Actual Price | Actual Quantity × Standard Price | Standard Quantity × Standard Price

Price Variance | Quantity Variance

Actual price is the amount actually paid for the for the input used.

A General Model for Variance Analysis

Actual Quantity × Actual Price | Actual Quantity × Standard Price | Standard Quantity × Standard Price

Price Variance | Quantity Variance

Standard price is the amount that should have been paid for the input used.

A General Model for Variance Analysis

Actual Quantity × Actual Price | Actual Quantity × Standard Price | Standard Quantity × Standard Price

Price Variance | Quantity Variance

(AQ × AP) – (AQ × SP) | (AQ × SP) – (SQ × SP)

AQ = Actual Quantity | SP = Standard Price
AP = Actual Price | SQ = Standard Quantity

Material Variances Example

Glacier Peak Outfitters has the following direct material standard for the fiberfill in its mountain parka.

0.1 kg. of fiberfill per parka at $5.00 per kg.

Last month 210 kgs of fiberfill were purchased and used to make 2,000 parkas. The material cost a total of $1,029.

Material Variances Summary

Actual Quantity × Actual Price	Actual Quantity × Standard Price	Standard Quantity × Standard Price
210 kgs. × $4.90 per kg.	210 kgs. × $5.00 per kg.	200 kgs. × $5.00 per kg.
= $1,029	= $1,050	= $1,000

Price variance $21 favorable

Quantity variance $50 unfavorable

Material Variances Summary

Actual Quantity × Actual Price	Actual Quantity × Standard Price	Standard Quantity × Standard Price
210 kgs. × $4.90 per kg.	210 kgs.	200 kgs. × $5.00 per kg.
= $1,029	= $1,050	= $1,000

$1,029 ÷ 210 kgs = $4.90 per kg

Price variance $21 favorable

Quantity variance $50 unfavorable

Material Variances Summary

Actual Quantity × Actual Price	Actual Quantity × Standard Price	Standard Quantity × Standard Price
210 kgs. × $4.90 per	210 kgs.	200 kgs. × $5.00 per kg.
= $1,029	= $1,050	= $1,000

0.1 kg per parka × 2,000 parkas = 200 kgs

Price variance $21 favorable

Quantity variance $50 unfavorable

Material Variances: Using the Factored Equations

Materials price variance

MPV = AQ (AP - SP)

= 210 kgs ($4.90/kg - $5.00/kg)

= 210 kgs (-$0.10/kg)

= $21 F

Materials quantity variance

MQV = SP (AQ - SQ)

= $5.00/kg (210 kgs-(0.1 kg/parka× 2,000 parkas))

= $5.00/kg (210 kgs - 200 kgs)

= $5.00/kg (10 kgs)

= $50 U

Isolation of Material Variances

I need the price variance sooner so that I can better identify purchasing problems.

You accountants just don't understand the problems that purchasing managers have.

I'll start computing the price variance when material is purchased rather than when it's used.

Material Variances

Hanson purchased and used 1,700 pounds. How are the variances computed if the amount purchased differs from the amount used?

The price variance is computed on the entire quantity purchased.

The quantity variance is computed only on the quantity used.

Responsibility for Material Variances

Materials Quantity Variance

Materials Price Variance

The standard price is used to compute the quantity variance so that the production manager is not held responsible for the purchasing manager's performance.

Responsibility for Material Variances

Quick Check ✓

Hanson Inc. has the following direct material standard to manufacture one Zippy:

1.5 pounds per Zippy at $4.00 per pound

Last week 1,700 pounds of material were purchased and used to make 1,000 Zippies. The material cost a total of $6,630.

Quick Check ✓

Hanson's material price variance (MPV) for the week was:

a. $170 unfavorable.
b. $170 favorable.
c. $800 unfavorable.
d. $800 favorable.

Quick Check ✓

Hanson's material price variance (MPV) for the week was:

a. $170 unfavorable.
b. $170 favorable.
c. $800 unfavorable.
d. $800 favorable.

MPV = AQ(AP - SP)
MPV = 1,700 lbs. × ($3.90 - 4.00)
MPV = $170 Favorable

Quick Check ✓

Hanson's material quantity variance (MQV) for the week was:

a. $170 unfavorable.
b. $170 favorable.
c. $800 unfavorable.
d. $800 favorable.

Quick Check ✓

Hanson's material quantity variance (MQV) for the week was:

a. $170 unfavorable.
b. $170 favorable.
c. $800 unfavorable.
d. $800 favorable.

MQV = SP(AQ - SQ)
MQV = $4.00(1,700 lbs - 1,500 lbs)
MQV = $800 unfavorable

Quick Check ✓

Actual Quantity × Actual Price	Actual Quantity × Standard Price	Standard Quantity × Standard Price
1,700 lbs. × $3.90 per lb.	1,700 lbs. × $4.00 per lb.	1,500 lbs. × $4.00 per lb.
= $6,630	= $ 6,800	= $6,000

Price variance $170 favorable

Quantity variance $800 unfavorable

Quick Check ✓ Continued

Hanson Inc. has the following material standard to manufacture one Zippy:

1.5 pounds per Zippy at $4.00 per pound

Last week 2,800 pounds of material were purchased at a total cost of $10,920, and 1,700 pounds were used to make 1,000 Zippies.

Quick Check ✓ Continued

Actual Quantity Purchased × Actual Price	Actual Quantity Purchased × Standard Price
2,800 lbs. × $3.90 per lb.	2,800 lbs. × $4.00 per lb.
= $10,920	= $11,200

Price variance
$280 favorable

Price variance increases because quantity purchased increases.

Quick Check ✓ Continued

Actual Quantity Used × Standard Price	Standard Quantity × Standard Price
1,700 lbs. × $4.00 per lb.	1,500 lbs. × $4.00 per lb.
= $6,800	= $6,000

Quantity variance is unchanged because actual and standard quantities are unchanged.

Quantity variance
$800 unfavorable

Labor Variances Example

Glacier Peak Outfitters has the following direct labor standard for its mountain parka.

1.2 standard hours per parka at $10.00 per hour

Last month employees actually worked 2,500 hours at a total labor cost of $26,250 to make 2,000 parkas.

Labor Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × $10.50 per hour	2,500 hours × $10.00 per hour.	2,400 hours × $10.00 per hour
= $26,250	= $25,000	= $24,000

Rate variance
$1,250 unfavorable

Efficiency variance
$1,000 unfavorable

Labor Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × $10.50 per hour	2,500 hours × $10.00 per hour	2,400 hours × $10.00 per hour
= $26,250	= $25,000	= $24,000

$26,250 ÷ 2,500 hours = $10.50 per hour

Rate variance
$1,250 unfavorable

Efficiency variance
$1,000 unfavorable

Labor Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × $10.50 per h	2,500 hours × $10.00 per hour	2,400 hours × $10.00 per hour
= $26,250	= $25,000	= $24,000

1.2 hours per parka × 2,000 parkas = 2,400 hours

Rate variance
$1,250 unfavorable

Efficiency variance
$1,000 unfavorable

Labor Variances: Using the Factored Equations

Labor rate variance

LRV = AH (AR - SR)
= 2,500 hours ($10.50 per hour – $10.00 per hour)
= 2,500 hours ($0.50 per hour)
= $1,250 unfavorable

Labor efficiency variance

LEV = SR (AH - SH)
= $10.00 per hour (2,500 hours – 2,400 hours)
= $10.00 per hour (100 hours)
= $1,000 unfavorable

Responsibility for Labor Variances

Production managers are usually held accountable for labor variances because they can influence the:

- Mix of skill levels assigned to work tasks.
- Level of employee motivation.
- Quality of production supervision.
- Quality of training provided to employees.

Responsibility for Labor Variances

Quick Check ✓

Hanson Inc. has the following direct labor standard to manufacture one Zippy:

1.5 standard hours per Zippy at $12.00 per direct labor hour

Last week 1,550 direct labor hours were worked at a total labor cost of $18,910 to make 1,000 Zippies.

Quick Check ✓

Hanson's labor rate variance (LRV) for the week was:

a. $310 unfavorable.
b. $310 favorable.
c. $300 unfavorable.
d. $300 favorable.

Quick Check ✓

Hanson's labor rate variance (LRV) for the week was:

a. $310 unfavorable.
b. $310 favorable.
c. $300 unfavo
d. $300 favorab

LRV = AH(AR - SR)
LRV = 1,550 hrs($12.20 - $12.00)
LRV = $310 unfavorable

Quick Check ✓

Hanson's labor efficiency variance (LEV) for the week was:

a. $590 unfavorable.

b. $590 favorable.

c. $600 unfavorable.

d. $600 favorable.

Quick Check ✓

Hanson's labor efficiency variance (LEV) for the week was:

a. $590 unfavorable.

b. $590 favorable.

c. $600 unfavorable.

d. $600 favorable.

LEV = SR(AH - SH)
LEV = $12.00(1,550 hrs - 1,500 hrs)
LEV = $600 unfavorable

Quick Check ✓

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
1,550 hours × $12.20 per hour	1,550 hours × $12.00 per hour	1,500 hours × $12.00 per hour
= $18,910	= $18,600	= $18,000

Rate variance
$310 unfavorable

Efficiency variance
$600 unfavorable

Variable Manufacturing Overhead Variances Example

Glacier Peak Outfitters has the following direct variable manufacturing overhead labor standard for its mountain parka.

1.2 standard hours per parka at $4.00 per hour

Last month employees actually worked 2,500 hours to make 2,000 parkas. Actual variable manufacturing overhead for the month was $10,500.

Variable Manufacturing Overhead Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × $4.20 per hour	2,500 hours × $4.00 per hour	2,400 hours × $4.00 per hour
= $10,500	= $10,000	= $9,600

Spending variance $500 unfavorable

Efficiency variance $400 unfavorable

Variable Manufacturing Overhead Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × $4.20 per hour	2,500 hours × $4.00 per hour	2,400 hours × $4.00 per hour
= $10,500	= $10,000	= $9,600

$10,500 ÷ 2,500 hours = $4.20 per hour

Spending variance $500 unfavorable

Efficiency variance $400 unfavorable

Variable Manufacturing Overhead Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × $4.20 per h	2,500 hours × $4.00 per hour	2,400 hours × $4.00 per hour
= $10,500	= $10,000	= $9,600

1.2 hours per parka × 2,000 parkas = 2,400 hours

Spending variance $500 unfavorable

Efficiency variance $400 unfavorable

Variable Manufacturing Overhead Variances: Using Factored Equations

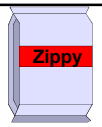
Variable manufacturing overhead spending variance

VMSV = AH (AR - SR)
= 2,500 hours ($4.20 per hour – $4.00 per hour)
= 2,500 hours ($0.20 per hour)
= $500 unfavorable

Variable manufacturing overhead efficiency variance

VMEV = SR (AH - SH)
= $4.00 per hour (2,500 hours – 2,400 hours)
= $4.00 per hour (100 hours)
= $400 unfavorable

Quick Check ✓



Hanson Inc. has the following variable manufacturing overhead standard to manufacture one Zippy:

1.5 standard hours per Zippy at $3.00 per direct labor hour

Last week 1,550 hours were worked to make 1,000 Zippies, and $5,115 was spent for variable manufacturing overhead.

Quick Check ✓

Hanson's spending variance (VOSV) for variable manufacturing overhead for the week was:

a. $465 unfavorable.
b. $400 favorable.
c. $335 unfavorable.
d. $300 favorable.

Quick Check ✓

Hanson's spending variance (VOSV) for variable manufacturing overhead for the week was:

a. $465 unfavorable.
b. $400 favorable.
c. $335 unfavora
d. $300 favorable

VOSV = AH(AR - SR)
VOSV = 1,550 hrs($3.30 - $3.00)
VOSV = $465 unfavorable

Quick Check ✓

Hanson's efficiency variance (VOEV) for variable manufacturing overhead for the week was:

a. $435 unfavorable.
b. $435 favorable.
c. $150 unfavorable.
d. $150 favorable.

Quick Check ✓

Hanson's efficiency variance (VOEV) for variable manufacturing overhead for the week was:

a. $435 unfavorable.
b. $435 favorable.
c. $150 unfavorable.
d. $150 favo

1,000 units × 1.5 hrs per unit

VOEV = SR(AH - SH)
VOEV = $3.00(1,550 hrs - 1,500 hrs)
VOEV = $150 unfavorable

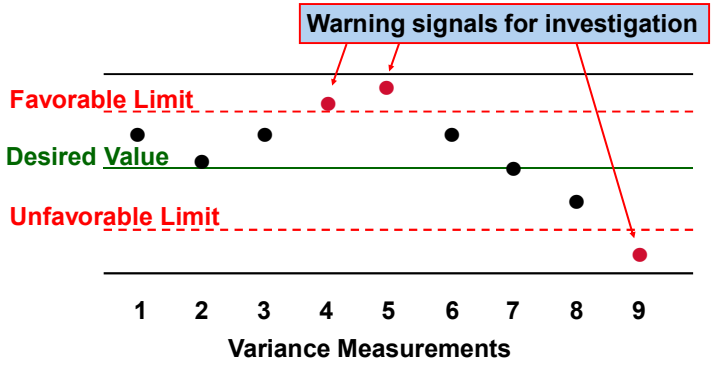
Quick Check ✓

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
1,550 hours × $3.30 per hour	1,550 hours × $3.00 per hour	1,500 hours × $3.00 per hour
= $5,115	= $4,650	= $4,500

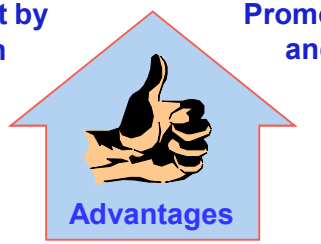
Spending variance $465 unfavorable

Efficiency variance $150 unfavorable

A Statistical Control Chart



Advantages of Standard Costs



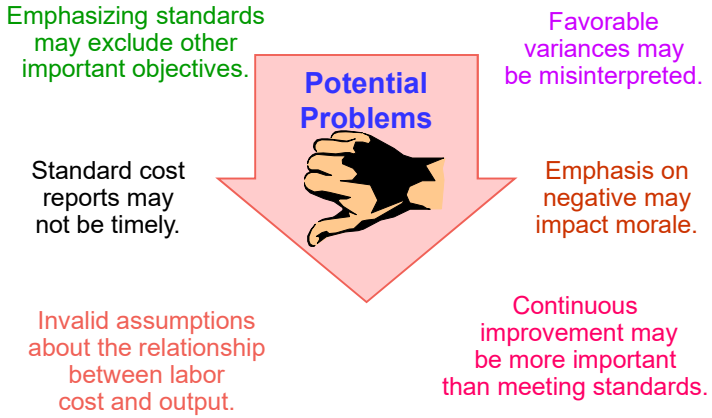
Management by exception

Promotes economy and efficiency

Simplified bookkeeping

Enhances responsibility accounting

Potential Problems with Standard Costs



End of Chapter 6